



Islamic University of Technology
Department of Computer Science and Engineering

Lab 2: Sorting

CSE 4404: Algorithms Lab(SWE)
Summer 2024-25

Contents

Task A. The Great Bottle Fill-Off **2**

Task Statement 2

Input Format 2

Output Format 2

Examples 2

Notes 2

Task B. Weighted Difference Minimization **3**

Task Statement 3

Input Format 3

Output Format 3

Examples 3

Task C. Count Values in Range **4**

Task Statement 4

Input Format 4

Output Format 4

Examples 4

Notes 4

Task A. The Great Bottle Fill-Off

Time Limit: 1 second | Memory Limit: 256 MB

You are given a container filled with a limited amount of water. The container can hold at most x liters of water. You are also given n empty bottles, each with a certain capacity. You want to fill as many bottles as possible using the water in the container, such that each bottle is filled to its full capacity. Your task is to determine the maximum number of bottles you can completely fill using the water from the container.

Input Format

The first line contains a single integer t ($1 \leq t \leq 20$) — the number of test cases.

For each test case:

- The first line contains two integers n and x ($1 \leq n \leq 10^7$, $1 \leq x \leq 10^{15}$) — the number of bottles and the total capacity of the container.
- The second line contains n space-separated integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^9$) — the amount of water required to completely fill each bottle.

The sum of n over all test cases does not exceed 10^7 .

Output Format

For each test case, print a single integer — the maximum number of bottles you can completely fill.

Examples

Sample Input	Sample Output
3	4
5 10	2
2 3 4 5 1	0
4 7	
5 3 4 2	
3 5	
6 7 8	

Notes

Comparison based sorting algorithms will give TLE because sorting up to 10^7 integers requires at least $\mathcal{O}(n \log n)$ comparisons, which is too slow under the given time limit.

Note that the values of a_i can be as large as 10^9 . Use proper non-comparison based sorting.

Task B. Weighted Difference Minimization

Time Limit: 1 second | Memory Limit: 256 MB

You are given three arrays a , b , and w , each containing n integers.

You have to rearrange the elements of all three arrays in any order so that the following value is minimized:

$$\sum_{i=1}^n w_i \cdot |a_i - b_i|$$

That is, after permuting the arrays, choose an arrangement such that the weighted sum of absolute differences between corresponding elements of a and b is as small as possible.

Input Format

The first line contains a single integer t ($1 \leq t \leq 10^4$) — the number of test cases.

For each test case:

- The first line contains a single integer n ($1 \leq n \leq 2 \times 10^5$) — the size of the arrays.
- The second line contains n integers $a_1, a_2, \dots, a_n$ ($|a_i| \leq 10^9$).
- The third line contains n integers $b_1, b_2, \dots, b_n$ ($|b_i| \leq 10^9$).
- The fourth line contains n integers $w_1, w_2, \dots, w_n$ ($1 \leq w_i \leq 10^9$).

It is guaranteed that the sum of n over all test cases does not exceed 2×10^5 .

Output Format

For each test case, print a single integer — the minimum possible value of $\sum_{i=1}^n w_i \cdot |a_i - b_i|$ after optimal rearrangement.

Examples

Sample Input	Sample Output
3	25
3	0
1 7 10	95
2 8 20	
5 1 10	
4	
1 2 3 4	
4 3 2 1	
1 1 1 1	
5	
10 10 10 10 10	
1 2 3 4 5	
5 4 3 2 1	

Task C. Count Values in Range

Time Limit: 2 seconds | Memory Limit: 512 MB

You are given an array of n integers. You also have to answer q queries.

For each query, you are given two integers l and r , and you must determine how many indices i ($1 \leq i \leq n$) satisfy the condition $l \leq a_i \leq r$.

Input Format

The first line contains two space-separated integers n and q ($1 \leq n, q \leq 10^5$) — the size of the array and the number of queries.

The second line contains n space-separated integers $a_1, a_2, \dots, a_n$ ($-10^9 \leq a_i \leq 10^9$) — the elements of the array.

Each of the next q lines contains two space-separated integers l and r ($-10^9 \leq l \leq r \leq 10^9$) — describing a query.

Output Format

For each query, print a single integer — the number of indices i such that $l \leq a_i \leq r$.

Examples

Sample Input	Sample Output
5 3	4
1 5 2 7 3	3
1 5	0
2 6	
8 10	

Notes

In the first query, the indices satisfying $1 \leq a_i \leq 5$ are $i = 1, 2, 3$, and 5 — a total of 4 indices.

In the second query, the indices satisfying $2 \leq a_i \leq 6$ are $i = 2, 3$, and 5 — a total of 3 indices.

In the third query, no index satisfies $8 \leq a_i \leq 10$.

Marks Distribution

Task	Marks
Task A	33.33%
Task B	33.33%
Task C	33.33%

Practice Problems

Problem	Links
Game	Codeforces: https://codeforces.com/contest/984/problem/A Vjudge: https://vjudge.net/problem/CodeForces-984A
Gravity Flip	Codeforces: https://codeforces.com/contest/405/problem/A Vjudge: https://vjudge.net/problem/CodeForces-405A
Valid Anagram	LeetCode: https://leetcode.com/problems/valid-anagram/
Jagged Swaps	Codeforces: https://codeforces.com/contest/1896/problem/A Vjudge: https://vjudge.net/problem/CodeForces-1896A
Unforgivable Curse (Easy)	Codeforces: https://codeforces.com/contest/1800/problem/E1 Vjudge: https://vjudge.net/problem/CodeForces-1800E1
Unforgivable Curse (Hard)	Codeforces: https://codeforces.com/contest/1800/problem/E2 Vjudge: https://vjudge.net/problem/CodeForces-1800E2